

Exact Bregman splitting without an attained solution

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Conjecture 00000008836. The assertion that splitting under Bregman distances always converges is false without a solvability hypothesis. Both source versions state unconditional convergence. The following example is an exact Bregman proximal method, hence the zero-error special case of a relative-error method.

Work on the real Hilbert space $\mathbb{R}$, with the smooth mirror function $h(x) = x^2/2$ and the proper closed convex objective $f(x) = -x$. The second split term can be the zero function. The actual mirror gradient is $\nabla h(x) = x$, and its Bregman distance is

$$D_h(y, x) = h(y) - h(x) - \langle \nabla h(x), y - x \rangle = \frac{(y - x)^2}{2}.$$

Thus $D_h \geq 0$, with equality exactly when $y = x$. The mirror has strong-convexity modulus one, since its defining first-order lower bound holds with equality.

The genuine unit-penalty proximal subproblem is

$$\min_y Q_x(y), \quad Q_x(y) = f(y) + D_h(y, x) = -y + \frac{(y - x)^2}{2}.$$

For every current state x and every competitor z ,

$$Q_x(z) - Q_x(x + 1) = \frac{(z - (x + 1))^2}{2}.$$

Consequently the unique global minimizer is $y = x + 1$. All subproblems are well-defined and uniquely solvable, despite the lack of a minimizer of the original objective.

The actual convex subgradient relation is $\partial f(y) = \{-1\}$ at every y : its definition is the inequality $v(z - y) \leq f(z) - f(y)$ for every z . Choosing $z = y \pm 1$ forces $v = -1$, which conversely satisfies every inequality. This graph is maximal monotone. Indeed any monotone extension containing a point (a, b) must be monotone against $(a - 1, -1)$ and $(a + 1, -1)$; these give $b \geq -1$ and $b \leq -1$.

At $y = x + 1$, choose $v = -1$ and enlargement error $e = 0$. Then the exact Bregman inclusion holds:

$$v + \nabla h(y) - \nabla h(x) = 0, \quad v \in \partial f(y).$$

In this Euclidean mirror, the usual relative-error certificate is also satisfied with zero residual, for every admissible $0 \leq \sigma < 1$:

$$v \in \partial_e f(y), \quad e \geq 0, \quad |v + y - x|^2 + 2e \leq \sigma^2 |y - x|^2.$$

Here $\partial_e f(y)$ is defined by $v(z - y) - e \leq f(z) - f(y)$ for all z . Its left-hand residual is zero. The associated exact proximal update is precisely $x_{k+1} = x_k + 1$.

For every starting point, induction gives $x_k = x_0 + k$, which has no finite limit. If it converged, its consecutive difference would converge to zero, whereas $x_{k+1} - x_k = 1$. Weak convergence also fails, as the identity is a continuous linear functional on this real Hilbert space.

Scope. The objective has no minimizer: $f(x + 1) < f(x)$ for every x . Its subgradient has no zero. This is the missing solvability hypothesis in the unconditional claim. We do not refute Bregman convergence theorems that assume a solution exists, and do not separately assess the stated summability characterization.

Formal verification. Lean proves the actual proper closed convex functions, mirror gradient and Bregman distance, full subgradient relation and maximality, the complete proximal argmin relation, the zero-relative-error inclusion, all iterates, and failure of norm and weak convergence for every starting point.